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United States Patent
Kind Code
Date of Patent
Inventor(s)

12412294
B2
September 09, 2025
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System, a detection system for detecting a foreign object on a runway and a method of the system

Abstract

The present invention relates to a system for detecting a foreign object on a runway. The system includes a thermal camera having a first field of view and adapted to capture a thermal image of an area of interest on the runway, a visible light camera having a second field of view and adapted to capture a visible light image of the area of interest on the runway, such that the first field of view overlaps the second field of view. The system further includes a processor configured to detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, and determine that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light image respectively.

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Family ID:	80951935
Appl. No.:	18/030056
Filed (or PCT Filed):	October 01, 2021
PCT No.:	PCT/SG2021/050598
PCT Pub. No.:	WO2022/071891
PCT Pub. Date:	April 07, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230360247 A1	Nov. 09, 2023

Foreign Application Priority Data

SG

10202009789R

Oct. 01, 2020

Publication Classification

Int. Cl.: **G06K9/00** (20220101); **G06T7/11** (20170101); **G06T7/62** (20170101); **G06T7/70** (20170101); **G06V10/10** (20220101); **G06V10/25** (20220101); **G06V20/52** (20220101); **G08G5/21** (20250101); **G08G5/54** (20250101); **H04N23/11** (20230101); **H04N23/23** (20230101)

U.S. Cl.:

CPC **G06T7/70** (20170101); **G06T7/11** (20170101); **G06T7/62** (20170101); **G06V10/16** (20220101); **G06V10/25** (20220101); **G06V20/52** (20220101); **G08G5/21** (20250101); **G08G5/54** (20250101); **H04N23/11** (20230101); **H04N23/23** (20230101); **G06T2207/10048** (20130101); **G06T2207/30232** (20130101); **G06V2201/07** (20220101)

Field of Classification Search

CPC: **G06T** (2207/30232); **G06T** (2207/10048); **G06T** (7/11); **G06T** (7/62); **G06T** (7/70); **G06T** (17/00); **G06T** (2200/08); **G06T** (2207/10012); **G06T** (2207/10024); **G06T** (2207/10028); **G06T** (2207/20021); **G06T** (2207/30244); **G06T** (7/12); **G06T** (7/593); **G06V** (20/52); **G06V** (2201/07); **G06V** (10/22)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a National Phase Application under 35 USC 371 of International Application No. PCT/SG2021/050598 filed Oct. 1, 2021, which claims priority to Singapore Application No. 10202009789R filed Oct. 1, 2020, all of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(2) The present invention relates to a system, a detection system for detecting a foreign object on a runway and a method of the system.

BACKGROUND

(3) Foreign objects and debris (FOD) on an airport runway pose a hazard to aircraft landing and taking-off thereon. There are FOD detection systems using visible light spectrum cameras to perform reliable FOD detection under normal clear weather conditions. Under normal clear weather conditions, e.g. in the absence of any fog, the FOD detection system will be able to capture high resolution images of any FOD and process them for the detection of the FOD with high accuracy. FOD may include engine and aircraft parts, tools, construction debris, rubber materials, natural materials, etc.

(4) However, during adverse weather conditions, especially under foggy weather conditions, the performance of the FOD detection system could be adversely impacted and compromised. The FOD detection system may not be able to reliably detect an FOD under foggy weather conditions, i.e. poor visibility conditions, as it operates only in the visible light spectrum. Hence, it will not be able to "see" the FOD under such conditions, e.g. through fog, which typically reduces visibility along the runway to less than 1 km. The visibility conditions may be categorised into different categories. For example, Cat II represents standard operations with associated Runway Visual Range (RVR) ranging from 550 m (1,800 feet) to 300 m (1,000 feet), Cat IIIa represents a precision instrument approach and landing operation with RVR not less than 175 m (600 feet), Cat IIIb represents a precision instrument approach and landing operation with RVR less than 175 m (600 feet) but not less than 50 m (200 feet), Cat IIIc represents a precision instrument approach and landing operation with no RVR limitations, i.e. even zero visibility. Depending on the geographical

location of the airports, the visibility of the runways at airports may vary and are categorised accordingly. While most FOD detection systems are able to detect FOD for airports with Cat II visibility, they are not able to be used for airport that experience Cat IIIa, Cat IIIb and Cat IIIc visibility.

(5) Further, the FOD detection system often generates invalid alerts or false positive alarms. The invalid alerts may be due to some phenomena, mainly light reflections, e.g. from artificial light sources originating from nearby buildings or runway edge lights, etc. These artificial lights reflecting off the smooth runway surface or reflecting off water puddles or ponding on the runway surface may cause the FOD detection system to identify it as a FOD and hence result in invalid alerts. The number of such invalid alerts due to reflections would typically increase significantly after a rainfall when water puddles or ponding are prevalent on the runway pavement surface. Though such reflections do occur during the daytime, they are much more prevalent at night and during the periods of dawn and dusk of the day.

(6) Therefore, it is important to provide a solution that enables the detection of FOD during poor visibility conditions, e.g. adverse weather conditions, and prevent or minimise false detection of the FOD.

SUMMARY

(7) According to various embodiments, a method for detecting a foreign object on a runway is provided. The method includes capturing a thermal image of an area of interest on the runway, capturing a visible light image of the area of interest on the runway, detecting a thermal object image in the thermal image, detecting a visible light object image in the visible light image, and determining that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively.

(8) According to various embodiments, determining the foreign object may include generating at least one attribute of the foreign object in each of the thermal object image and visible light object image, comparing the at least one attribute of the foreign object in the thermal object image and the visible light object image, such that the foreign object is detected when the at least one attribute of the foreign object in thermal object image and the visible light object image are the same.

(9) According to various embodiments, the at least one attribute of the foreign object may include the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image.

(10) According to various embodiments, the at least one attribute of the foreign object in thermal object image and the visible light object image are the same when the distance between the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image is within a position parameter.

(11) According to various embodiments, the at least one attribute of the foreign object may include the size of the thermal object image and visible light object image.

(12) According to various embodiments, the at least one attribute of the foreign object in the thermal object image and visible light object image are the same when the difference in the size of the thermal object image in the thermal image and the size of the visible light object image in the visible light image is within a size parameter.

(13) According to various embodiments, the method may further include obtaining an enlarged thermal object image and an enlarged visible light object image when the foreign object is detected.

(14) According to various embodiments, the method may further include identifying an object category of the foreign object in the thermal image, such that identifying the object category may include, segmenting the thermal image into a plurality of thermal image regions, assigning a feature vector to each of the plurality of thermal image regions, comparing the feature vector to a plurality of reference feature vectors, such that each of the plurality of reference feature vectors represents an object category, identifying the reference feature vector closest to the feature vector and its object category.

- (15) According to various embodiments, segmenting the thermal image may include labelling each pixel in the thermal image and grouping the labelled pixels with the same characteristic into a plurality of groups to form the plurality of thermal image regions.
- (16) According to various embodiments, the method may further include identifying an object category of the foreign object in the visible light image, such that identifying the object category may include, segmenting the visible light image into a plurality of visible light image regions, assigning a feature vector to each of the plurality of visible light image regions, comparing the feature vector to a plurality of reference feature vectors, such that each of the plurality of reference feature vectors represents an object category, and identifying the reference feature vector closest to the feature vector and its object category.
- (17) According to various embodiments, segmenting the visible light image may include labelling each pixel in the visible light image and grouping the labelled pixels with the same characteristic into a plurality of groups to form the plurality of visible light image regions.
- (18) According to various embodiments, the method may further include training the thermal camera to detect the foreign object based on the visible light images from the visible light camera.
- (19) According to various embodiments, training the thermal camera may include determining a relationship between the object category of the foreign object in the visible light image and the temperature of the foreign object in the thermal image.
- (20) According to various embodiments, a system for detecting a foreign object on a runway is provided. The system includes a thermal camera having a first field of view and adapted to capture a thermal image of an area of interest on the runway, a visible light camera having a second field of view and adapted to capture a visible light image of the area of interest on the runway, such that the first field of view overlaps the second field of view. The system further includes a processor in communication with the thermal camera and the visible light camera, a memory in communication with the processor for storing instructions executable by the processor, such that the processor is configured to detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, and determine that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively.
- (21) According to various embodiments, to determine the foreign object, the processor may be configured to generate at least one attribute of the foreign object in each of the thermal object image and visible light object image, compare the at least one attribute of the foreign object in the thermal object image and the visible light object image, such that the foreign object is detected when the at least one attribute of the foreign object in thermal object image and the visible light object image are the same.
- (22) According to various embodiments, the at least one attribute of the foreign object may include the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image.
- (23) According to various embodiments, the at least one attribute of the foreign object in thermal object image and the visible light object image are the same when the distance between the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image is within a position parameter.
- (24) According to various embodiments, the at least one attribute of the foreign object may include the size of the thermal object image and visible light object image.
- (25) According to various embodiments, the at least one attribute of the foreign object in the thermal object image and visible light object image are the same when the difference in the size of the thermal object image in the thermal image and the size of the visible light object image in the visible light image is within a size parameter.
- (26) According to various embodiments, the processor may be further configured to zoom in the thermal camera and visible light camera to obtain an enlarged thermal object image and an enlarged

visible light object image when the foreign object is detected.

(27) According to various embodiments, the processor may be configured to identify an object category of the foreign object in the thermal image, such that the processor may be configured to, segment the thermal image into a plurality of thermal image regions, assign a feature vector to each of the plurality of thermal image regions, compare the feature vector to a plurality of reference feature vectors, such that each of the plurality of reference feature vectors represents an object category, identify the reference feature vector closest to the feature vector and its object category.

(28) According to various embodiments, to segment the thermal image, the processor may be configured to label each pixel in the thermal image and group the labelled pixels with the same characteristic into a plurality of groups to form the plurality of thermal image regions.

(29) According to various embodiments, the processor may be further configured to identify an object category of the foreign object in the visible light image, such that the processor may be configured to, segment the visible light image into a plurality of visible light image regions, assign a feature vector to each of the plurality of visible light image regions, compare the feature vector to a plurality of reference feature vectors, such that each of the plurality of reference feature vectors represents an object category, identify the reference feature vector closest to the feature vector and its object category.

(30) According to various embodiments, to segment the visible light image, the processor may be configured to label each pixel in the visible light image and group the labelled pixels with the same characteristic into a plurality of groups to form the plurality of visible light image regions.

(31) According to various embodiments, the processor may be further configured to train the thermal camera to detect the foreign object based on the visible light images from the visible light camera.

(32) According to various embodiments, to train the thermal camera, the processor may be configured to determine a relationship between the object category of the foreign object in the visible light image and the temperature of the foreign object in the thermal image.

(33) According to various embodiments, a detection system for detecting a foreign object on a runway divided into a plurality of sectors is provided, the detection system includes a plurality of sets of cameras spaced apart from each other, each of the plurality of sets of cameras may include a thermal camera includes a first field of view and adapted to capture a thermal image of an area of interest on the runway, a visible light camera includes a second field of view and adapted to capture a visible light image of the area of interest on the runway, such that the first field of view overlaps the second field of view, a processor in communication with the thermal camera and the visible light camera, a memory in communication with the processor for storing instructions executable by the processor, such that the processor may be configured to detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, and determine that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively, such that each of the plurality of sets of cameras may be configured to scan one of the plurality of sectors of the runway.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 shows a schematic diagram of an exemplary embodiment of a system for detecting a foreign object on a runway.

(2) FIG. 1A shows a schematic diagram of the visible light image with the visible light object image of the foreign object therein and the thermal image with the thermal object image of the foreign object therein.

(3) FIG. 2A shows an exemplary embodiment of the system.

- (4) FIG. 2B shows the system in FIG. 2A scanning one of the plurality of sectors of the runway.
- (5) FIG. 3 shows an exemplary embodiment of a detection system for detecting a foreign object on a runway divided into a plurality of sectors.
- (6) FIG. 4 shows a flow diagram of an exemplary method for detecting a foreign object on a runway.
- (7) FIG. 5 shows a flow diagram of an exemplary method for detecting the foreign object on a runway.
- (8) FIG. 6 shows a flow diagram of an exemplary method for detecting the foreign object on a runway.
- (9) FIG. 7 shows a flow diagram of an exemplary method for comparing the at least one attribute of the foreign object in the thermal object image and the visible light object image.
- (10) FIG. 8 shows a flow diagram of a method for identifying a foreign object on the runway.
- (11) FIG. 9 shows a flow diagram of a method for identifying a foreign object on the runway.
- (12) FIG. 10 shows a flow diagram of a method for training the image identification module to improve the identification of a foreign object on the runway.
- (13) FIG. 11 shows a flow diagram of a method for detecting a foreign object with a thermal camera.

DETAILED DESCRIPTION

- (14) In the following examples, reference will be made to the figures, in which identical features are designated with like numerals.
- (15) FIG. 1 shows a schematic diagram of an exemplary embodiment of a system **100** for detecting a foreign object **20** on a runway. System **100** includes a thermal camera **110** having a first field of view **110F** and adapted to capture a thermal image **110M** of an area of interest **112** on the runway, a visible light camera **120** having a second field of view **120F** and adapted to capture a visible light image of the area of interest **112** on the runway, such that the first field of view **110F** overlaps the second field of view **120F**, a processor **132** in communication with the thermal camera **110M** and the visible light camera **120M**, a memory **134** in communication with the processor **132** for storing instructions executable by the processor **132**, such that the processor **132** is configured to detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, and determine that the foreign object **20** is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively. System **100** may include a server comprising the processor **132**, the memory **134**, an I/O interface **136** configured to provide an interface between the processor **132** and peripheral interface modules, e.g. keyboard, mouse, touchscreen, display, etc. System **100** may include a communication module **138** configured to facilitate communication, wired or wirelessly, between the system **100** and other user devices, e.g. mobile devices, laptops, via the internet. System **100** may include a storage device **140** configured to store data. System **100** may include a display, e.g. monitor, touchscreen, for displaying signals, e.g. alert signal, to the operator. System **100** is configured to detect a foreign object, debris (FOD), on a runway, a taxiway, aprons, ramps, etc. under both day and night ambient light condition without assisted illumination, e.g. visible spectrum illumination, infrared illumination, laser illumination.
- (16) FIG. 1A shows a schematic diagram of the visible light image **120M** with the visible light object image **120B** of the foreign object **20** therein and the thermal image **110M** with the thermal object image **110B** of the foreign object **20** therein.
- (17) System **100** may include an image processing module **134M** (see FIG. 1) configured to process images **110T**, **120T** captured from the thermal camera **110** and visible light camera **120**. System **100** may include a thermal camera operating module **134T** containing operating parameters of the thermal camera **110** for operating the thermal camera **110**. System **100** may include a visible light camera operating module **134V** containing operating parameters of the visible light camera **120** for operating the visible light camera **120**. The modules **134T**, **134V**, **134M** may be stored in

the storage device **140** and loaded into the memory **134** to be processed by the processor **132**.

(18) Upon capturing the thermal image **110M** and visible light image **120M**, the images **110M**, **120M** may be transmitted to the processor **132** to be processed. Processor **132** may receive and process the thermal image **110M** and the visible light image **120M** to detect a foreign object **20** on the runway. System **100** enables detection of a foreign object **20** for airports with Cat II visibility, Cat IIIa visibility, Cat IIIb visibility and Cat IIIc visibility, such that it enables the detection of foreign object during poor visibility conditions, e.g. adverse weather conditions, and prevent or minimise false detection of the foreign object.

(19) FIG. 2A shows an exemplary embodiment of the system **200**. System **200** may include a set of cameras, i.e. the visible light camera **220** and the thermal camera **210**. The set of cameras **210S** may be mounted rigidly on an actuator **250** adapted to move the set of cameras **210S**. Set of cameras **210S** may be controlled by the processor **132** to scan a sector of the runway to detect the foreign object **20** on the surface of the runway.

(20) Actuator **250** may be a pan and tilt unit (PTU) adapted to pan and tilt the set of cameras **210S** simultaneously so that the set of cameras **210S** are able to have the same field of view and focus on the same area of interest. Actuator **250** may be adapted to pan the set of cameras **210S** in the horizontal direction **210H** and/or tilt the set of cameras **210S** in the vertical direction **210V**.

Actuator **250** may be in communication with the processor **132** such that the processor **132** may be configured to remotely control the movement of the actuator **250** to pan and tilt the set of cameras **210S** to scan the runway. Actuator **250** may be installed on top of a support **252**, e.g. a mast structure, which is typically located along the runway. Support may be located at distance of 120 m-350 m from the centreline **304** (see FIG. 3) of the runway.

(21) FIG. 2B shows the system **100** in FIG. 2A scanning one of the plurality of sectors **202S** of the runway **202**. Each set of cameras **210S** may include the thermal camera **210** and the visible light camera **220**, each having a field of view **210F**, **220F** and is adapted to capture an area of interest **212**. Field of view **210F** of the visible light camera **220** may overlap with the field of view **220F** of the thermal camera **210**. Both the field of views **210F**, **220F** of the visible light camera **220** and the thermal camera **210** may cover a specific area of interest **212** within the sector **202S** of the runway **202**. Hence, as a result of the overlapping field of views **210F**, **220F**, both the visible light camera **220** and the thermal camera **210** may detect the same foreign object **20** on the sector **202S** of the runway at the same time while scanning the sector **202S**.

(22) FIG. 3 shows an exemplary embodiment of a detection system **300** for detecting a foreign object **20** on a runway **302** divided into a plurality of sectors **302S**. Detection system **300** includes a plurality of sets of cameras **3105** spaced apart from each other. Each of the plurality of sets of cameras **3105** includes a thermal camera **210** having a first field of view **210F** and adapted to capture a thermal image **110M** of an area of interest **212** on the runway **302**, a visible light camera **220** having a second field of view **220F** and adapted to capture a visible light image **120M** of area of interest **212** on the runway **302**, such that the first field of view **210F** overlaps the second field of view **220F**. Detection system **300** further includes a processor, a memory in communication with the processor for storing instructions executable by the processor, such that the processor is configured to detect a thermal object image **110B** in the thermal image **110M**, detect a visible light object image **120B** in the visible light image **120M**, and determine that the foreign object **20** is detected when the thermal object image **110B** and the visible light object image **120B** are detected in the thermal image **110M** and the visible light object image **120B** respectively, such that each of the plurality of a set of cameras **3105** is configured to scan one of the plurality of sectors **302S** of the runway **302**. As shown in FIG. 3, the runway **302** may be divided into a plurality of sectors **302S**. Each of the plurality of sets of cameras **3105** may scan one of the plurality of sectors **302S** to detect any foreign object **20** on the surface of the respective sector **302S**. Each of the plurality of sectors **302S** may be further divided into a plurality of subsectors. Each set of cameras **3105** may be operable to scan a dedicated sector **302S** and scan the sector **302S**, subsector by subsector. In

this way, when a foreign object **20** is detected, the system **300** is able to identify the sector **302S** based on the set of cameras **3105** scanning the sector **302S**. Set of cameras **3105** may scan the sector **302S** in a specific typical scan direction, e.g. from the left-most subsector to the right-most subsector or the right-most subsector to the left-most subsector.

(23) Thermal camera **210** detects foreign object **20** on the runway **302** by detecting the difference in thermal radiation level (or temperature) between the foreground, i.e. foreign object **20**, and the background, i.e. the runway surface. Thermal camera **210** operates in the infrared spectrum and does not require any ambient light to enable it to “see” the foreign object **20**. Thermal camera **210** may also be commonly known as infrared thermal camera. Thermal camera **210** may be a Mid Wave Infrared (MWIR) camera or a Long Wave Infrared (LWIR) camera. Thermal camera **210** provides the advantage to detect the foreign object **20** on the runway **302** under very low visibility conditions and even under zero illumination conditions, i.e. total darkness. Hence, the thermal camera **210** provides the advantage of the ability to detect the foreign object **20** on the runway **302** even under foggy weather conditions. Thermal camera **210** may capture and transmit images and video output in monochrome to the processor **132**. Thermal camera **210** is entirely passive with no active transmissions or emissions, e.g. radio frequency, microwave, artificial illumination, infrared, laser and LIDAR etc. As such, the thermal camera **210** offers the following advantages, e.g. no interference with existing airport systems/equipment and aircraft systems/equipment, no interference with future airport systems/equipment and aircraft systems/equipment, no licensing and approval of frequency/spectrum required from airport and frequency spectrum regulator.

(24) Unlike the thermal camera **210**, the visible light camera **220** operates within the visible spectrum of light and hence requires some minimum amount of ambient visible spectrum light to enable it to “see” the foreign object **20** on the runway **302**. Visible light camera **220** is not able to detect any foreign object **20** when the visibility conditions are too poor or under zero illumination conditions. For example, the visible light camera **220** is also not able to detect the foreign object **20** when the visibility condition (above the runway surface) is very poor or in the presence of fog (above the runway surface). Visible light camera **220** is able to capture and transmit full colour and high-resolution images/video, e.g. Full HD (FHD) or 4K Ultra HD (4K UHD) resolution. The colour images in high resolution enables reliable and accurate visual verification and confirmation of the detected foreign object **20** by an operator, as well as reliable and accurate recognition/classification of the detected foreign object **20** by the system **300**. Therefore, the combined use of both the visible light camera **220** and the thermal camera **210** enables the system **300** to operate under very low visibility conditions, e.g. foggy weather conditions, to enable the system **300** to detect foreign object **20** on the runway **302** surface accurately and reliably. Visible light camera **220** is configured to capture and output visible light image **120M** in colour and high resolution to the processor **132**. Visible light camera **220** does not require any transmission of infrared illumination, visible spectrum illumination or laser illumination to operate. Being passive, the system **300** provides the advantage that it does not pose any hazard or cause any interference to other airport systems and/or aircraft systems, e.g. for aircraft landing/taking-off from the runway **302**. System **300** provides the following advantages, no interference with existing airport systems/equipment and aircraft systems/equipment, no interference with future airport systems/equipment and aircraft systems/equipment, no licensing and approval of frequency/spectrum required from airport and frequency spectrum regulator.

(25) FIG. 4 shows a flow diagram of an exemplary method **1000** for detecting a foreign object **20** on a runway. Method includes capturing a thermal image **110M** of an area of interest **112** on the runway in block **1010**, capturing a visible light image **120M** of the area of interest **112** on the runway in block **1020**, detecting a thermal object image **110B** in the thermal image **110M** in block **1030**, detecting a visible light object image **120B** in the visible light image **120M** in block **1040**, and determining that the foreign object **20** is detected when the thermal object image **110B** and the visible light object image **120B** are detected in the thermal image **110M** and the visible light object

image **120B** respectively in block **1050**. Thermal object image **110B** may be a portion of the thermal image **110M** representing the foreign object **20** in the thermal image **110M** and may be casually known as the foreign object **20** in the thermal image **110M**. Visible light object image **120B** may be a portion of the visible light image **120M** representing the foreign object **20** in the visible light image **120M** and may be casually known as the foreign object **20** in the visible light image **120M**.

(26) Before capturing the thermal image **110M** and visible light image **120M**, the method may include scanning the runway with the thermal camera **110** and the visible light camera **120**. As the thermal camera **110** and the visible light camera **120** scan a sector of the runway, the thermal camera **110** and the visible light camera **120** captures thermal images **110M** and visible light images **120M** of a plurality of area of interests **112** along the sector. To detect the foreign object **20**, the image processing module **134M** may process the thermal image **110M** and the visible light image **120M** to determine if the foreign object **20** is present in the thermal image **110M** and the visible light image **120M**. Upon detecting the foreign object **20**, the image processing module **134M** may be configured to identify the thermal object image **110B** and the visible light object image **120B** within the thermal image **110M** and the visible light image **120M** respectively. Upon identifying the foreign object **20**, the system **100** may generate an alert signal.

(27) To detect the foreign object **20**, the method may include generating at least one attribute of the foreign object **20** in each of the thermal object image **110B** and visible light object image **120B**, comparing the at least one attribute of the foreign object **20** in the thermal object image **110B** and the visible light object image **120B**, such that the foreign object **20** is detected when the at least one attribute of the foreign object **20** in thermal object image **110B** and the visible light object image **120B** are the same or within a specified parameter or threshold level. System **100** may be configured to obtain an enlarged thermal object image **110B** and an enlarged visible light object image **120B** when the foreign object **20** is detected by zooming the visible light camera **120** and thermal camera **110** onto the detected foreign object **20**.

(28) FIG. 5 shows a flow diagram of an exemplary method **2000** for detecting the foreign object **20** on a runway. System **100** may be configured to designate the visible light camera **120** to be a primary detector and the thermal camera **110** to be a secondary detector. Referring to FIG. 5, in block **2110**, the visible light camera **120** may be configured to scan one of the plurality of sectors on the runway. Visible light camera **120** may be configured to scan a subsector by subsector of the sector. Visible light camera **120** may capture a plurality of visible light images **120M** within each sector. Plurality of visible light images **120M** may be processed by the image processing module **134M** to detect the foreign object **20**. In block **2210**, the thermal camera **110** may be configured to scan the same sector on the runway scanned by the visible light camera **120**. Thermal camera **110** may be configured to scan a subsector by subsector of the sector. Thermal camera **110** may capture a plurality of thermal image **110M**s within the same sector. Plurality of thermal images **110M** may be processed by the image processing module **134M** to detect the foreign object **20**. Thermal camera **110** and the visible light camera **120** may be configured to scan the sector concurrently. Image processing module **134M** for the visible image and thermal image **110M** may be separate modules for processing the visible light image **120M** and thermal image **110M** respectively.

(29) In block **2120**, the system **100** may detect a foreign object **20** after processing the visible light image **120M**. System **100** may identify the visible light object image **120B** within the visible image. In block **2220**, the system **100** may detect a foreign object **20** after processing the thermal image **110M**. System **100** may identify the thermal object image **110B** within the thermal image **110M**. Thermal image **110M** and the visible light image **120M** may be processed by the processor **132** concurrently. If the system **100** detects a foreign object **20** in the visible light image **120M**, the system **100** may generate a “Suspected FOD” alert signal to inform the operator that a foreign object **20** has been detected in the visible light image **120M** in block **2130**. Similarly, if the system **100** detects a foreign object **20** in the thermal image **110M**, the system **100** may generate a

“Suspected FOD” alert signal to inform the operator that a foreign object **20** has been detected in the thermal image **110M** in block **2230** as the detection of the foreign object **20** has yet to be verified. The “Suspected FOD” signal may be generated for each of the visible light image **120M** and the thermal image **110M**. System **100** may display the thermal object image **110B** and/or the visible light object image **120B** on the display for the operator to view. System **100** may generate at least one attribute of the visible light object image **120B** and of the thermal object image **110B**. At least one attribute may include the position of the visible light object image **120B** in the visible light image **120M**, the position of the thermal object image **110B** in the thermal image **110M**, the size of the visible light object image **120B** and/or the size of the thermal object image **110B**. For example, the system **100** may generate the position of the visible light object image **120B** in the visible light image **120M** and the position of the thermal object image **110B** in the thermal image **110M** and/or the size of the visible light object image **120B** and thermal object image **110B**. In block **2140**, the system **100** may be configured to determine whether the foreign object **20** is detected in the visible light image **120M** and the thermal image **110M** by comparing the at least one attribute of the visible light object image **120B** and the thermal object image **110B**. Details of this comparing step may be shown in FIG. 7. If the attributes of the visible light object image **120B** and the thermal object image **110B** matches in block **2150**, the system **100** determines that the foreign object **20** is detected in the visible light image **120M** and the thermal image **110M**. System **100** may receive an operator verification input via the peripheral interface module to verify the detection of the foreign object **20** after viewing the visible light image **120M** and/or thermal image **110M** on the display. System **100** may identify the foreign object **20** based on the at least one attributes in block **2160**. Once the foreign object **20** has been detected and/or identified, the system **100** may generate an alert signal, e.g. a “Confirmed FOD” signal in block **2170**. Otherwise, the system **100** may generate a “No Confirmed FOD” alert signal. System **100** may transmit the alert signal to an operator's mobile device or display the alert signal on the display for the operator's viewing. System **100** may generate the alert signal upon receiving the operator verification input that the foreign object **20** is detected.

(30) FIG. 6 shows a flow diagram of an exemplary method **3000** for detecting the foreign object **20** on a runway. Method **3000** is identical to method **2000** in FIG. 5 except that the system **100** is configured to designate the thermal camera **110** to be a primary detector and the visible light camera **120** to be a secondary detector. Like reference numerals in FIG. 5 and FIG. 6 represent the same steps. Referring to FIG. 6, in block **3110**, the thermal camera **110** may be configured to scan one of the plurality of sectors on the runway. Thermal camera **110** may be configured to scan a subsector by subsector of the sector. Thermal camera **110** may capture a plurality of thermal images **110M** within each sector. Plurality of thermal images **110M** may be processed by the image processing module **134M** to detect the foreign object **20**. In block **3210**, the visible light camera **120** may be configured to scan the same sector on the runway scanned by the thermal camera **110**. Visible light camera **120** may be configured to scan a subsector by subsector of the sector. Visible light camera **120** may capture a plurality of visible light images **120M** within the sector. Plurality of visible light images **120M** may be processed by the image processing module **134M** to detect the foreign object **20**. Thermal camera **110** and the visible light camera **120** may be configured to scan the sector concurrently. In block **3120**, the system **100** may detect a foreign object **20** after processing the thermal image **110M**. System **100** may identify the thermal object image **110B** within the thermal image **110M**. In block **3230**, the system **100** may detect a foreign object **20** after processing the visible light image **120M**. System **100** may identify the visible light object image **120B** within the visible light image **120M**. Thermal image **110M** and the visible light image **120M** may be processed by the processor **132** concurrently. Blocks **3140** to **3170** are identical to blocks **2140** to **2170** in FIG. 5.

(31) FIG. 7 shows a flow diagram of an exemplary method **4140** for comparing the at least one attribute of the foreign object **20** in the thermal object image **110B** and the visible light object

image **120B**. Method **4140** is used in method **2000** in FIG. 5 and method **3000** in FIG. 6 in block **2140** and block **3140** respectively. At least one attribute of the foreign object **20** may include the position of the thermal object image **110B** in the thermal image **110M** and the position of the visible light object image **120B** in the visible light image **120M**. In block **4141**, the at least one attribute of the foreign object **20** in thermal object image **110B** and the visible light object image **120B** may be considered the same when the distance between the position of the thermal object image **110B** in the thermal image **110M** and the position of the visible light object image **120B** in the visible light image **120M** is within a position parameter. For example, the processor **132** identifies the positional difference between the positions of the visible light object image **120B** and thermal object image **110B** in the visible light image **120M** and the thermal image **110M** and determines if the positional difference between the positions are within a position parameter, i.e. a pre-defined position threshold level. Position parameter may be determined based on statistical analysis of the detected positions of all the detected foreign object **20** samples. If the positional difference is within the position parameter, the processor **132** may generate a “position match” alert signal in block **4142**.

(32) At least one attribute of the foreign object **20** may include the size of the thermal object image **110B** and visible light object image **120B**. In block **4143**, the at least one attribute of the foreign object **20** in the thermal object image **110B** and visible light object image **120B** may be considered the same when the difference in the size of the thermal object image **110B** in the thermal image **110M** and the size of the visible light object image **120B** in the visible light image **120M** is within a size parameter. For example, the processor **132** identifies the size difference between the sizes of the visible light object image **120B** and thermal object image **110B** in the visible light image **120M** and the thermal image **110M** and determines if the size difference between the positions are within a size parameter, i.e. a pre-defined size threshold level. Size parameter may be determined based on statistical analysis of the measured sizes of all the detected foreign object **20** samples. If the size difference is within the size parameter, the processor **132** may generate a “size match” alert signal in block **4144**.

(33) Depending on the configuration of the system **100**, the process may detect the foreign object **20** based on the position and/or size of the thermal object image **110B** and visible light object image **120B**. For example, where both the position and size of the thermal object image **110B** and the visible light object image **120B** are used, the foreign object **20** is detected when the position and size of the thermal object image **110B** and visible light object image **120B** are within the position parameter and size parameter respectively, i.e. matched. System **100** may generate an alert signal when the foreign object **20** is detected, e.g. generate an “attribute match” signal when the attributes are matched in block **4145**. System **100** may generate the alert signal when the “position match” alert signal and “size match” alert signal is on or generated.

(34) The exemplary system **100** and methods described above provide a solution that enables the detection of a foreign object **20** during adverse weather conditions and prevent or minimise false detection of the foreign object **20**. For example, the reflections from water puddles or ponding after a rainfall, or the reflections on a smooth runway surface occur within the visible spectrum of light. As, the visible light camera **120** operates solely within the visible light spectrum, the system **100** may easily misinterpret these reflections as foreign objects **20**, or “Suspected FOD”. This would cause the system **100** to generate an invalid alert or false positive alarms. Therefore, by comparing and detecting the foreign object **20** using both the thermal image **110M** and visible light image **120M**, the system **100** is able to provide a more accurate detection of the foreign object **20** and prevent or minimise false detection of the foreign object **20**.

(35) Referring to the method **2000** in FIG. 5 and method **3000** in FIG. 6. When comparing the at least one attribute of the visible light object image **120B** and the thermal object image **110B**, the system **100** may determine that the detection of the foreign object **20** is not confirmed, i.e. the at least one attribute in the visible light object image **120B** does not match the at least one attribute in

the thermal image **110M**. System **100** may then generate a “No Confirmed FOD” alert signal. In this situation, as the system **100** “suspected” that a foreign object **20** is detected, but identified that it is not “confirmed”, the system **100** may be configured to identify this event as an invalid alert or false positive alert. Hence, the system **100** may be configured to store at least one of: the alert signal, attributes, features and the images of this event into a database, e.g. invalid alert database, for post-mortem analysis and investigation.

(36) FIG. **8** shows a flow diagram of a method **5000** for identifying a foreign object **20** on the runway. Image processing module **134M** may be configured to carry out the method **5000**. System **100** may store a plurality of reference feature vectors and an object category associated with each of the plurality of reference feature vectors in a reference feature vector database, which may be stored in the storage device **140**. To identify the foreign object **20**, the system **100** is configured to identify the object category of the foreign object **20**. Referring to FIG. **8**, the method may include capturing the visible light image **120M** and the thermal image **110M** via the visible light camera **120** and the thermal camera **110** in block **5302**. To identify an object category of the foreign object **20** in the thermal image **110M**, the method includes segmenting the thermal image **110M** into a plurality of thermal image regions in block **5304**. Method may include segmenting the visible light image **120M** into a plurality of visible light image regions in block **5304**. Segmenting the thermal image **110M** may include labelling each pixel in the thermal image **110M** and grouping the labelled pixels with the same characteristic into a plurality of groups to form the plurality of thermal image regions. Segmenting the visible light image **120M** may include labelling each pixel in the visible light image **120M** and grouping the labelled pixels with the same characteristic into a plurality of groups to form the plurality of visible light image regions. Processor **132** may be configured to assign a label to each pixel in the thermal image **110M** and visible light image **120M** such that the pixels with the same label share certain common characteristics or properties. Upon segmentation, the thermal image **110M** and the visible light image **120M** are made up of a plurality of thermal image regions and a plurality of visible light image regions that collectively cover the respective images. The pixels within each of the plurality of regions are similar with respect to some characteristic, feature or property, such as texture, colour, or intensity. Adjacent regions of the plurality of image regions are significantly different from each other with respect to the same characteristic. The segmented thermal image **110M** and visible light image **120M** may be used to detect and locate region or regions which may include the suspected foreign object **20** in the images. In block **5306**, the method may include detecting and extracting features from the thermal image **110M** and visible light image **120M**. System **100** may be configured to assign a feature vector to each of the plurality of thermal image regions, e.g. thermal feature vector, and each of the plurality of visible light image regions, e.g. visible light feature vector. A feature may refer to a pattern or distinct structure found in an image, such as a point, blob, small patch, corner, edge. Features are represented by an image region which differs from the image regions in its immediate surroundings, e.g. by texture, colour or intensity. Features may be extracted, grouped and represented by the feature vector. A foreign object **20** may be represented by a group of features, which may be represented by a feature vector.

(37) In block **5308**, the method may include comparing the feature vector to the plurality of reference feature vectors, such that each of the plurality of reference feature vectors is associated to an object category. System **100** may match the feature vector to the plurality of reference feature vectors in block **5308**. Each object category, e.g. rubber tire, mechanic's tool, aircraft part, vehicle part etc., may be represented by a specific reference feature vector stored in the reference feature vector database. Each extracted feature vector may be matched against the plurality of reference feature vectors in the reference feature vector database. In block **5310**, the method may include detecting the foreign object **20**. If there is a match between the feature vector and one or more of the plurality of reference feature vectors, the system **100** may determine that a foreign object **20** is detected. System **100** may generate a “Suspected FOD” alert signal. In block **2312**, the method may

include identifying the object category of the foreign object **20**. System **100** may be configured to identify the reference feature vector closest to the feature vector and its object category. Based on the matched one or more of the plurality of reference feature vectors, the system **100** may identify or classify the foreign object **20** based on the closest match between the feature vector and the one or more of the plurality of reference feature vectors, e.g. the “shortest distance” between the feature vector and the specific reference feature vector. In addition, the “shortest distance” may be used to determine the match or probability of the foreign object **20** being classified accurately. There could potentially be more than one reference feature vector which may match the feature vector. The matching may be based on fuzzy matching. System **100** may be configured to recognise and classify the foreign object **20** based on the object category. System **100** may identify an object category of the foreign object **20** in the visible light image **120M**. Based on the matched reference feature vector, the object category tagged to the matched reference feature vector may be retrieved and the foreign object **20** may be identified or classified. Upon identifying the foreign object **20**, the system **100** may generate and transmit an alert signal.

(38) FIG. **9** shows a flow diagram of a method **6000** for identifying a foreign object **20** on the runway. Method **6000** is identical to method **5000** in FIG. **8** except that the system **100** is configured to automatically detect and extract the features from the thermal image **110M** and visible light image **120M**, match the feature vector to the plurality of reference feature vectors and detecting the foreign object **20** in block **6306**. Like reference numerals in FIG. **8** and FIG. **9** represent the same steps. System **100** may be configured to train the image processing module **134M** using deep learning modules and automatically carry out the steps in block **6306**.

(39) Thermal camera **110** is able to detect foreign objects **20** by detecting the difference in the temperature, i.e. the infrared thermal radiation of the foreground, e.g. the foreign object **20**, with respect to the background, e.g. the runway **202** surface. Different categories or types of foreign objects **20** are made of different materials, e.g. metallic, rubber, plastic, concrete, etc., and would have different energy absorptivity, reflectivity and emissivity. As such, different categories of foreign objects **20** would result in different temperature, i.e. different level of infrared thermal radiation with respect to the background, i.e. the runway surface. The difference in temperature between the foreign object **20** and the runway would be detectable by the thermal camera **110**.

(40) Therefore, it is beneficial to “train” the thermal camera **110**, or rather the thermal camera operating module **134T**, to differentiate the different categories of foreign objects **20** by identifying the type of material which the foreign object **20** is made of, e.g. rubber, metallic, plastic, concrete, asphalt, etc. As foreign objects **20** made of the different types of materials would have different emissivity resulting in different level of temperature and different temperature contrast level with respect to the background, i.e. the runway, the “well-trained” thermal camera **110** would be able to identify a foreign object **20** more accurately.

(41) To train the thermal camera **110**. Under normal clear weather conditions, the thermal camera **110** may be put through an initial period of “training” whereby the thermal camera **110** may operate in “training” mode to enable it to “learn” from the visible light images **120M** of the visible light camera **120**. After the initial “training”, the thermal camera **110** may be adequately “learned” to enable the thermal camera **110** to provide reliable and accurate foreign object **20** detection with relatively high level of accuracy. With a high level of accuracy, it would then be possible to enable a system **100** with a “standalone” thermal camera **110** instead of a set of visible light camera **120** and thermal camera **110**. In this way, the system **100** will be applicable under adverse weather conditions and/or very low visibility conditions without the visible light camera **120**.

(42) FIG. **10** shows a flow diagram of a method for training the image processing module **134M** to improve the identification of a foreign object **20** on runway. In block **7110**, the visible light camera **120** may be configured to scan one of the plurality of sectors on the runway. Visible light camera **120** may be configured to scan a subsector by subsector of the sector. Visible light camera **120** may capture a plurality of visible light images **120M** of the sector. Plurality of visible light images

120M may be processed by the image processing module **134M** to detect the foreign object **20**. In block **7210**, the thermal camera **110** may be configured to scan the same sector on the runway scanned by the visible light camera **120**. Thermal camera **110** may be configured to scan a subsector by subsector of the sector. Thermal camera **110** may capture a plurality of thermal images **110M** of the sector.

(43) Plurality of thermal images **110M** may be processed by the image processing module **134M** to detect the foreign object **20**. Thermal camera **110** and the visible light camera **120** may be configured to scan the sector concurrently. In block **7120**, the system **100** may detect a foreign object **20** after processing the visible light image **120M** and identify the visible light object image **120B**. In block **7220**, the system **100** may detect a foreign object **20** after processing the thermal image **110M** and identify the thermal object image **110B**. Thermal image **110M** and the visible light image **120M** may be processed by the processor **132** concurrently. If the system **100** detects a foreign object **20** in the visible light image **120M**, the system **100** may generate a “Suspected FOD” alert signal to inform the operator that a foreign object **20** has been detected in the visible light image **120M**. Similarly, if the system **100** detects a foreign object **20** in the thermal image **110M**, the system **100** may generate a “Suspected FOD” signal to inform the operator that a foreign object **20** has been detected in the thermal image **110M** in block **7230** as the detection of the foreign object **20** has yet to be verified. The “Suspected FOD” alert signal may be generated for each of the visible light image **120M** and the thermal image **110M**. System **100** may display the thermal object image **110B** and/or the visible light object image **120B** on the display for the operator to view. System **100** may generate at least one attribute of the visible light object image **120B** in block **7130** and generate at least one attribute of the thermal object image **110B** in block **7230**.

(44) At least one attribute may include the position of the visible light object image **120B** in the visible light image **120M**, the position of the thermal object image **110B** in the thermal image **110M**, the size of the visible light object image **120B**, the size of the thermal object image **110B** and/or the temperature of the thermal object image **110B**. For example, the system **100** may generate the position of the visible light object image **120B** in the visible light image **120M** and/or the size of the visible light object image **120B**. For example, the system **100** may generate at least one of the position of the thermal object image **110B** in the thermal image **110M**, the size of the thermal object image **110B** and the temperature of the foreign object **20**. System **100** may be configured to store at least one of: the alert signal, attributes, features and the images of this event into the foreign object alert signal and event database **742** in block **7132** and block **7232**. In block **7140**, the system **100** may be configured to determine whether the foreign object **20** is present in the visible light image **120M** and the thermal image **110M** by comparing the at least one attribute of the visible light object image **120B** and the thermal object image **110B**. The method of comparing the at least one attribute may be shown in method **4140** in FIG. 7. If the attributes of the visible light object image **120B** and the thermal object image **110B** matches in block **7150**, the system **100** determines that the foreign object **20** is detected in the visible light image **120M** and the thermal image **110M**. If the foreign object **20** has been detected, the system **100** may generate and display an alert signal, e.g. a “Confirmed FOD” alert signal in block **7170**. Otherwise, the system **100** may generate and transmit a “No Confirmed FOD” alert signal. System **100** may be configured to store at least one of: the signal, attributes, features and the images into the foreign object alert signal and event database **742** in block **7172**. In block **7180**, the system **100** may be configured to optimize detection configuration parameters of the thermal camera **110**. System **100** may optimize the detection configuration parameters based on the data stored in the foreign object alert signal and event database **742**. System **100** may run statistical analysis and/or optimization modules to optimise the detection configuration parameters based on the data. System **100** may use artificial intelligence to optimize the detection configuration parameters based on the data. In block **7182**, the system **100** may be configured to store the optimized detection configuration parameters of the thermal camera **110** in a foreign object **20** detection configuration parameters database **744**.

for the thermal camera **110**.

(45) System **100** may determine the relationship between the object category and the temperature of the foreign object **20** in the thermal image **110M** for all the detected/verified foreign object samples. Processor **132** may be further configured to train the thermal camera **110** to detect the foreign object **20** based on the visible light images **120M** from the visible light camera **120**. As it is substantially easier to identify and categorise a foreign object **20** in visible light image **120M**, the system **100** may form a relationship between the object category of the visible light object image **120B** obtained from the visible light camera **120** and the temperature of the thermal object image **110B** from the thermal camera **110**. Hence, with a sufficiently large foreign object sample size, the system **100** would be able to determine the relationship between different foreign object categories, e.g. foreign object made of different materials, such as metal, plastic, rubber, etc. and their corresponding temperatures. System **100** may then be able to build an “FOD Type Thermal Profile Model” which could be used to map the various foreign object types, i.e. made of different materials, to their corresponding temperature ranges. In this way, the system may be able to identify the foreign object **20** more easily based on the thermal object image **110B** thereof.

(46) The “FOD Type Thermal Profile Model” would enable the system **100** to determine the foreign object category or type, including the specific type of material which the foreign object **20** is made of, such as metal, rubber, plastic, etc. of any detected foreign object **20** based on the temperature of the foreign object **20** detected by the thermal camera **110**. The development of the “FOD Type Thermal Profile Model” may be based on mathematical methods and/or statistically methods, such as statistical correlation analysis. Alternatively, the development of the “FOD Type Thermal Profile Model” may be based on artificial intelligence and machine learning technologies. This “FOD Type Thermal Profile Model” may be used to optimize the detection configuration parameters of the thermal camera **110**.

(47) To optimize the performance of the thermal camera **110**, it is necessary to optimize the detection configuration parameters of the thermal camera **110**. The detection configuration parameters of the thermal camera **110** may be a set of operating parameters pertaining to the thermal camera **110** to enable the thermal camera **110** to detect foreign objects **20** with optimum and high level of accuracy. Operating parameters of the thermal camera **110** may include sensitivity, gain, brightness, contrast, shutter timing settings, etc. In this way, as the system **100** trains the thermal camera operating module **134T**, the detection performance of the thermal camera **110** would be improved over time to a level at which it may be able to operate as “standalone” and sole foreign object detector for the system **100**, i.e. without the visible light camera **120**. The optimized performance thermal camera **110** would be beneficial under adverse weather conditions and/or under very low visibility conditions.

(48) The various temperature contrast levels due to different types of foreign object materials may be detected by the thermal camera **110**. This would enable the thermal camera **110** to detect a foreign object **20** accurately. This may also enable the system **100** to classify or recognise the different categories or types of foreign object **20** based on the different types of materials which the foreign object **20** is made of.

(49) Database **742** may contain the alert signals, e.g. “Suspected FOD”, “Confirmed FOD” and events that took place in the methods for both the visible light camera **120** and the thermal camera **110**. Database **742** may store the detected and/or computed foreign object **20** attributes, e.g. category, size, position, temperature etc.

(50) FIG. **11** shows a flow diagram of a method **8000** for detecting a foreign object **20** with a thermal camera **110**. After the “training”, the thermal camera **110** may be able to provide reliable and accurate foreign object **20** detection with relatively high level of accuracy. With a high level of accuracy, it would then be possible for the thermal camera **110** to be used without the visible light camera **120**. Referring to FIG. **11**, the system **100** may be configured to detect a foreign object **20** by leveraging on optimized detection configuration parameters for the thermal camera **110** stored in

the detection configuration parameters database **844** for the thermal camera **110** in block **8402**. Thermal camera **110** may be configured as the sole foreign object detector to detect a foreign object **20**.

(51) A skilled person would appreciate that the features described in one example may not be restricted to that example and may be combined with any one of the other examples.

(52) The present invention relates to a system **100**, a detection system **100** for detecting a foreign object on a runway and a method of the system **100** generally as herein described, with reference to and/or illustrated in the accompanying drawings.

Claims

1. A method for identifying a foreign object on a runway based on a thermal object image of the foreign object, the method comprising: capturing a thermal image of an area of interest on the runway, capturing a visible light image of the area of interest on the runway, detecting a thermal object image in the thermal image, detecting a visible light object image in the visible light image, determining that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively, identifying and categorizing the visible light object image into an object category of the foreign object, determining a relationship between the object category of the foreign object in the visible light image and the thermal object image of the foreign object, wherein when a thermal object image of the foreign object is detected, the foreign object is identified by mapping the object category to the thermal object image based on the relationship.
2. The method according to claim 1, wherein determining the foreign object comprises generating at least one attribute of the foreign object in each of the thermal object image and visible light object image, comparing the at least one attribute of the foreign object in the thermal object image and the visible light object image, wherein the foreign object is detected when the at least one attribute of the foreign object in thermal object image and the visible light object image are the same.
3. The method according to claim 2, wherein the at least one attribute of the foreign object comprises the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image.
4. The method according to claim 3, wherein the at least one attribute of the foreign object in thermal object image and the visible light object image are the same when the distance between the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image is within a position parameter.
5. The method according to claim 2, wherein the at least one attribute of the foreign object comprises the size of the thermal object image and visible light object image.
6. The method according to claim 5, wherein the at least one attribute of the foreign object in the thermal object image and visible light object image are the same when the difference in the size of the thermal object image in the thermal image and the size of the visible light object image in the visible light image is within a size parameter.
7. The method according to claim 1, further comprising identifying an object category of the foreign object in the thermal image, wherein identifying the object category comprises, segmenting the thermal image into a plurality of thermal image regions, assigning a feature vector to each of the plurality of thermal image regions, comparing the feature vector to a plurality of reference feature vectors, wherein each of the plurality of reference feature vectors represents an object category, identifying the reference feature vector closest to the feature vector and its object category.
8. The method according to claim 7, wherein segmenting the thermal image comprises labelling each pixel in the thermal image and grouping the labelled pixels with the same characteristic into a

plurality of groups to form the plurality of thermal image regions.

9. The method according to claim 1, wherein identifying the object category comprises, segmenting the visible light image into a plurality of visible light image regions, assigning a feature vector to each of the plurality of visible light image regions, comparing the feature vector to a plurality of reference feature vectors, wherein each of the plurality of reference feature vectors represents an object category, and identifying the reference feature vector closest to the feature vector and its object category.

10. The method according to claim 9, wherein segmenting the visible light image comprises labelling each pixel in the visible light image and grouping the labelled pixels with the same characteristic into a plurality of groups to form the plurality of visible light image regions.

11. The method according to claim 1, further comprising generating a thermal profile model of the foreign object based on the relationship and optimizing the detection configuration parameters of the thermal camera based on the thermal profile model.

12. A detection system for identifying a foreign object on a runway based on a thermal object image of the foreign object, the system comprising: a thermal camera comprising a first field of view and adapted to capture a thermal image of an area of interest on the runway, a visible light camera comprising a second field of view and adapted to capture a visible light image of the area of interest on the runway, wherein the first field of view overlaps the second field of view, a processor in communication with the thermal camera and the visible light camera, a memory in communication with the processor for storing instructions executable by the processor, wherein the processor is configured to, detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, determine that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively identify and categorizing the visible light object image into an object category of the foreign object, determine a relationship between the object category of the foreign object in the visible light image and the thermal object image of the foreign object, wherein when a thermal object image of the foreign object is detected, the foreign object is identified by mapping the object category to the thermal object image based on the relationship.

13. The detection system according to claim 12, wherein to determine the foreign object, the processor is configured to generate at least one attribute of the foreign object in each of the thermal object image and visible light object image, compare the at least one attribute of the foreign object in the thermal object image and the visible light object image, wherein the foreign object is detected when the at least one attribute of the foreign object in thermal object image and the visible light object image are the same.

14. The detection system according to claim 13, wherein the at least one attribute of the foreign object comprises the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image.

15. The detection system according to claim 14, wherein the at least one attribute of the foreign object in thermal object image and the visible light object image are the same when the distance between the position of the thermal object image in the thermal image and the position of the visible light object image in the visible light image is within a position parameter.

16. The detection system according to claim 13, wherein the at least one attribute of the foreign object comprises the size of the thermal object image and visible light object image.

17. The detection system according to claim 16, wherein the at least one attribute of the foreign object in the thermal object image and visible light object image are the same when the difference in the size of the thermal object image in the thermal image and the size of the visible light object image in the visible light image is within a size parameter.

18. The detection system according to claim 12, wherein the processor is configured to identify an object category of the foreign object in the thermal image, wherein the processor is configured to, segment the thermal image into a plurality of thermal image regions, assign a feature vector to each

of the plurality of thermal image regions, compare the feature vector to a plurality of reference feature vectors, wherein each of the plurality of reference feature vectors represents an object category, identify the reference feature vector closest to the feature vector and its object category.

19. The detection system according to claim 18, wherein to segment the thermal image, the processor is configured to label each pixel in the thermal image and group the labelled pixels with the same characteristic into a plurality of groups to form the plurality of thermal image regions.

20. The detection system according to claim 12, wherein to identify an object category of the foreign object in the visible light image, the processor is configured to, segment the visible light image into a plurality of visible light image regions, assign a feature vector to each of the plurality of visible light image regions, compare the feature vector to a plurality of reference feature vectors, wherein each of the plurality of reference feature vectors represents an object category, identify the reference feature vector closest to the feature vector and its object category.

21. The detection system according to claim 20, wherein to segment the visible light image, the processor is configured to label each pixel in the visible light image and group the labelled pixels with the same characteristic into a plurality of groups to form the plurality of visible light image regions.

22. The detection system according to claim 12, wherein the process is further configured to generate a thermal profile model of the foreign object and optimizing the detection configuration parameters of the thermal camera based on the thermal profile model.

23. A system for detecting a foreign object on a runway divided into a plurality of sectors, the detection system comprising: a plurality of sets of cameras spaced apart from each other, each of the plurality of sets of cameras comprises: a thermal camera comprising a first field of view and adapted to capture a thermal image of an area of interest on the runway, a visible light camera comprising a second field of view and adapted to capture a visible light image of the area of interest on the runway, wherein the first field of view overlaps the second field of view, a processor in communication with the plurality of sets of cameras, a memory in communication with the processor for storing instructions executable by the processor, wherein the processor is configured to: detect a thermal object image in the thermal image, detect a visible light object image in the visible light image, and determine that the foreign object is detected when the thermal object image and the visible light object image are detected in the thermal image and the visible light object image respectively, identify and categorizing the visible light object image into an object category of the foreign object, determine a relationship between the object category of the foreign object in the visible light image and the thermal object image of the foreign object, wherein when a thermal object image of the foreign object is detected, the foreign object is identified by mapping the object category to the thermal object image based on the relationship, wherein each of the plurality of sets of cameras is configured to scan one of the plurality of sectors of the runway.
